

Exercise 3.5 Yaw Assensoh Opoku

1a

The screenshot displays a PostgreSQL client interface with the following components:

- Object Explorer:** A tree view on the left showing the database structure. It includes Servers (4), Databases (3), and Schemas (1). The 'Rockbuster' database is selected, showing its various components like Casts, Catalogs, and Schemas.
- Query Editor:** The central area where a SQL query is entered:

```
1 SELECT title, film_id
2 FROM film
3
```
- Data Output:** A table at the bottom showing the results of the query. It has two columns: 'title' (character varying (255)) and 'film_id' ([PK] integer). The results are as follows:

	title	film_id
1	Chamber Italian	133
2	Grosse Wonderful	384
3	Airport Pollock	8
4	Bright Encounters	98
5	Academy Dinosaur	1
6	Ace Goldfinger	2
- Status Bar:** At the bottom, it indicates 'Total rows: 1000', 'Query complete 00:00:00.091', and 'Ln 1, Col 22'.

1 **EXPLAIN**
2 **SELECT** title, film_id
3 **FROM** film
4

Data Output Messages Explain X Notifications

Showing rows: 1 to 1 Page No: 1 of 1

	QUERY PLAN text
1	Seq Scan on film (cost=0.00..98.00 rows=1000 width=19)

Total rows: 1 Query complete 00:00:00.077

Query Query History

1 **EXPLAIN**
2 **SELECT** *
3 **FROM** film
4

Data Output Messages Explain X Notifications

Showing rows: 1 to 1 Page No: 1 of 1

	QUERY PLAN text
1	Seq Scan on film (cost=0.00..98.00 rows=1000 width=384)

Total rows: 1 Query complete 00:00:00.073

1b.

1c. These two queries are basically the same with costs and rows but different from their width but the original query retrieves all columns whiles the revised query retrieved from title and film_id columns. For optimization I'd suggest using LIMIT instead.

2a

Rockbuster/postgres@PostgreSQL 17

Query Query History

```
1 SELECT title, release_year, rental_rate
2 FROM film
3 ORDER BY release_year DESC, rental_rate DESC, title ASC
```

Data Output Messages Explain X Notifications

Showing rows: 1 to 1000 Page 1

	title character varying (255)	release_year integer	rental_rate numeric (4,2)
1	Ace Goldfinger	2006	4.99
2	Airplane Sierra	2006	4.99
3	Airport Pollock	2006	4.99
4	Aladdin Calendar	2006	4.99
5	Ali Forever	2006	4.99
6	Amelie Hellfighters	2006	4.99
7	American Circus	2006	4.99
8	Anthem Luke	2006	4.99
9	Apache Divline	2006	4.99
10	Apocalypse Flamingos	2006	4.99
11	Attacks Hate	2006	4.99
12	Attraction Newton	2006	4.99
13	Autumn Crow	2006	4.99
14	Baby Hall	2006	4.99
15	Backlash Undefeated	2006	4.99
16	Beast Hunchback	2006	4.99
17	Beauty Grease	2006	4.99
18	Behavior Runaway	2006	4.99
19	Betrayed Rear	2006	4.99
20	Bikini Borrowers	2006	4.99
21	Bilko Anonymous	2006	4.99
22	Birch Antitrust	2006	4.99
23	Bird Independence	2006	4.99
24	Birds Perdition	2006	4.99
25	Blindness Gun	2006	4.99
26	Boiled Dares	2006	4.99
27	Boogie Amelie	2006	4.99
28	Born Spinal	2006	4.99
29	Bowfinger Gables	2006	4.99

Total rows: 1000 Query complete 00:00:00.102

3a

Query Query History

```
1 SELECT rating, AVG(rental_rate)
2 FROM film
3 GROUP BY rating
4
```

Data Output Messages Explain X Notifications

Showing rows: 1 to 5 Page No: 1 of 1

	rating mpaa_rating	avg numeric
1	PG	3.0518556701030928
2	PG-13	3.0348430493273543
3	NC-17	2.9709523809523810
4	G	2.8888764044943820
5	R	2.9387179487179487

Total rows: 5 Query complete 00:00:00.097

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Query Query History

```

1 SELECT rating, MAX(rental_duration), MIN(rental_duration)
2 FROM film
3 GROUP BY rating
4

```

Data Output Messages Explain × Notifications

Showing rows: 1 to 5 Page No: 1 of 1

	rating mpaa_rating	max smallint	min smallint
1	PG	7	3
2	PG-13	7	3
3	NC-17	7	3
4	G	7	3
5	R	7	3

3b.

4a.

This is the process of ETL- for the Data Engineer

-Extract, which involves collecting the data from multiple data source.

-Transform, the extracted data and convert it into another format.

-Load, where the transformed data is inserted into the new database which is Rockbuster's data base.

4b. There could be lots of problems if the data hasn't gone through the ETL process like missing or incomplete data, data duplications and could be a total waste of time.